

[Assignment-2]

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Section: 01

course
code : CSE 260

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Initial

Answer to the question no 1

$$\begin{aligned}& (A+B)(C+B)(C+A) \\&= (AC+AB+BC+BB)(C+A) \\&= ACC+ABC+BCC+BBC+AAC+AAB+AB C+AB B \\&= AC+ABC+BC+BC+AC+AB+ABC+AB \\&= AC+ABC+BC+AB \\&= AB(C+1)+AC+BC \\&= AB+AC+BC\end{aligned}$$

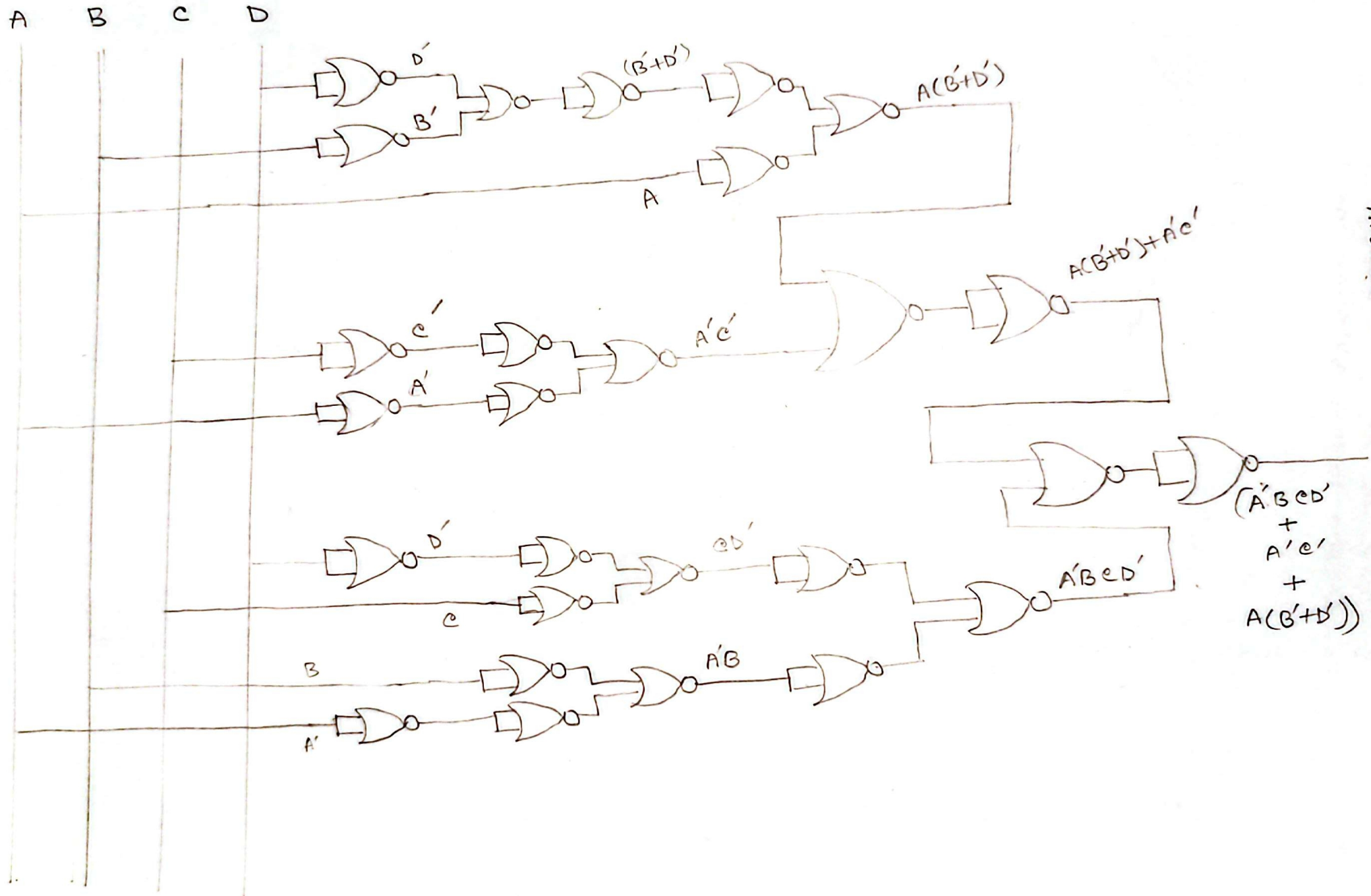
(Ans)

Answer to the question no 2

$$\begin{aligned}& (x+y'+z')(x'+y)(x'+z') \\&= (x \cdot y' \cdot z') + (x' \cdot y) + (x' \cdot z') \\&= (x' \cdot y \cdot z) + (x \cdot y') + (x \cdot z)\end{aligned}$$

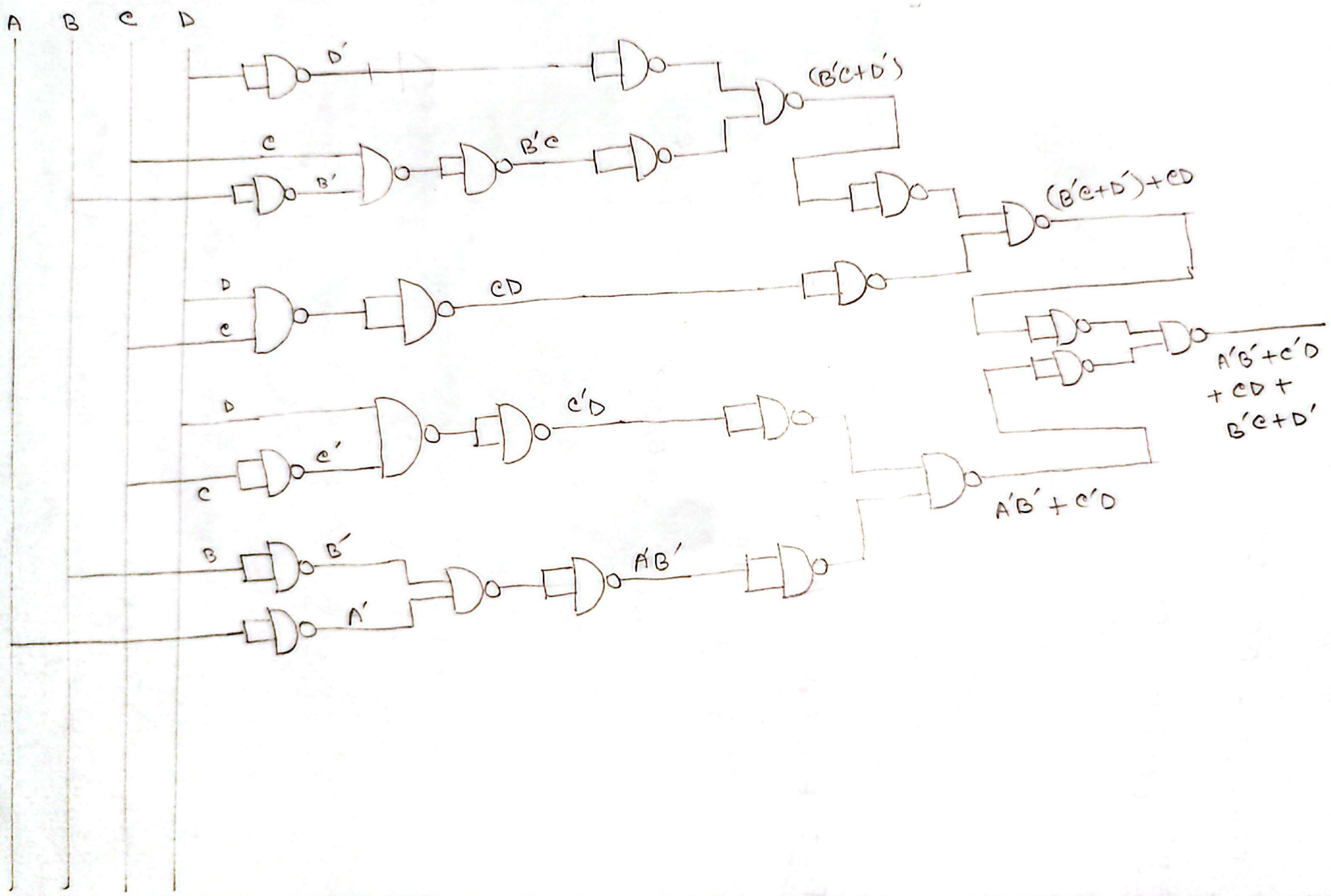
(Ans)

$$F(A, B, C, D) = (A'B'CD' + A'C' + A(B'+D'))$$



Answer to the question no 3

$$F(A,B,C,D) = (A'B' + c'D + cD + (B'e + D'))$$



Ans to the question no 4

Ans to the question no 5

$$F(A, C, B) = A'C + BC'$$

SOP:

$$F(A, C, B) = A'C + BC'$$

$$= A'C(B+B') + BC'(A+A')$$

$$= \underset{0}{A'} \underset{1}{C} B + \underset{0}{A'} \underset{1}{C} B' + \underset{1}{A} \underset{0}{C'} B + \underset{0}{A'} \underset{0}{C'} B$$

$$= \sum (3, 2, 5, 1)$$

$$= \sum (1, 2, 3, 5)$$

(Ans)

POS: $F(A, C, B) = \frac{A'C}{x} + \frac{BC'}{y \cdot z}$

$$= \left(\frac{A'C+B}{y \cdot z} \right) \left(\frac{A'C+C'}{y \cdot z} \right)$$

$$= (B+A')(B+C) (C'+A') (C'+C)$$

$$= (A'+B+C') (B+C+AA') (C'+A'+BB')$$

$$= (A'+B+C) (A'+B+C') (B+C+A) (B+C+A')$$

$$(A'+C'+B) (A'+C'+B')$$

$$= (A'+C+B) (A'+C'+B) (A+C+B) (A'+C+B)$$

$$(A'+C'+B) (A'+C'+B')$$

$$= (A' + C + B) (A' + C' + B) (A + C + B) (A' + C' + B')$$

$$\begin{matrix} & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{matrix}$$

$$= \pi(4, 6, 0, 7)$$

$$= \pi(0, 4, 6, 7)$$

Ans: SOP : $\Sigma(1, 2, 3, 5)$

POS : $\pi(0, 4, 6, 7)$

Ans to the question no 6

$$F(A, B, C, D) = A + B'C' + D$$

SOP:

$$F(A, B, C, D) = A + B'C' + D$$

$$= A(B+B')(C+C')(D+D') + B'C'(A+A')(D+D') + D(A+A')(B+B')(C+C')$$

$$= (AB + AB')(CD + CD' + C'D + C'D') + (B'C'A + B'C'A')(D + D') + (DA + DA')(BC + BC' + B'C + B'C')$$

$$= \begin{matrix} \text{ABED} & \text{ABED} & \text{ABC'D} & \text{ABC'D} & \text{AB'CD} & \text{AB'CD} \\ \text{1111} & \text{1110} & \text{1101} & \text{1100} & \text{1011} & \text{1010} \end{matrix}$$

$$+ \begin{matrix} \text{AB'C'D} & \text{AB'C'D} & \text{AB'C'D} & \text{AB'C'D} & \text{AB'C'D} & \text{AB'C'D} \\ \text{1001} & \text{1000} & \text{1001} & \text{1000} & \text{1001} & \text{1000} \end{matrix}$$

$$+ \begin{matrix} \text{A'B'C'D} & \text{A'B'C'D} & \text{A'B'C'D} & \text{A'B'C'D} & \text{A'B'C'D} & \text{A'B'C'D} \\ \text{0001} & \text{0000} & \text{0001} & \text{0000} & \text{0001} & \text{0000} \end{matrix}$$

$$+ \begin{matrix} \text{AB'ED} & \text{AB'ED} & \text{AB'ED} & \text{AB'ED} & \text{AB'ED} & \text{AB'ED} \\ \text{0111} & \text{0110} & \text{0101} & \text{0100} & \text{0011} & \text{0010} \end{matrix}$$

$$= \Sigma(15, 14, 13, 12, 11, 10, 9, 8, 1, 0, 7, 5, 3)$$

$$= \sum (0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

POS: $F(A, B, C, D) = \overline{(A'B' + C'D)} = A + B'C' + D$
 $= (A + D + B')(A + D + C')$

$x=0$
 $x'=1$

$$= (A + B' + C'C' + D) (A + B'B' + C' + D)$$

$$= \begin{pmatrix} A & B' & C & D \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} A & B' & C' & D \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} A & B & C' & D \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} A & B' & C' & D \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$= \prod (4, 6, 2)$$

$$= \prod (2, 4, 6)$$

Ans: SOP: $\sum (0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15)$

POS: $\prod (2, 4, 6)$

Ans to the question no 7

$$F(A, B, C, D, E) = ACE + CDE$$

SOP: $F(A, B, C, D, E) = ACE + CDE$
 $= ACE(B+B')(D+D') + CDE(A+A')(B+B')$

$x=1$
 $x=0$
 $= (ABCE + AB'CE)(D+D') + (ACDE + A'CDE)(B+B')$

$$= \begin{matrix} \text{ABCE} & + & \text{AB'CE} & + & \text{ACDE} & + & \text{A'CDE} \\ \text{1111} & & \text{1110} & & \text{1011} & & \text{1011} \\ \text{ABCE} & + & \text{AB'CE} & + & \text{ACDE} & + & \text{A'CDE} \\ & & \text{0111} & & \text{0011} & & \end{matrix}$$

$$= \sum (31, 29, 23, 21, 15, 7)$$

$$= \sum (7, 15, 21, 23, 29, 31)$$

16 8 4 2 1

POS: $F(A, B, C, D, E) = ACE + CDE$
 $= (ACE + CD)(ACE + E)$
 $= (ACE + C)(ACE + D)(AC + E)(E + E)$
 $= (C + AC)(C + E)(D + AC)(D + E)(E + A)(E + C)$
 $= (C + A)(C + C)(C + E)(D + A)(D + C)(D + E)(E + A)(E + C)$
 $= (C + A)(C + E)(D + A)(D + C)(D + E)(E + A)(E + C)$
 $= (C + A + BB' + DD' + EE')(A + BB' + C + DD' + E)$
 $(A + BB' + CC' + D + EE')(AA' + BB' + C + D + EE')$
 $(AA' + BB' + CC' + D + E)(A + BB' + CC' + DD' + E)$
 $(AA' + BB' + C + DD' + E)$

$$\begin{aligned}
&= (A + BB' + c + DD' + E)(A + BB' + c + DD' + E') \\
&\quad (AA' + BB' + c + D + E)(AA' + BB' + c + D' + E) \\
&\quad (A + BB' + cc' + D + E)(A + BB' + cc' + D + E') \\
&\quad (AA' + BB' + c + D + E)(AA' + BB' + c + D + E') \\
&\quad (AA' + BB' + c + D + E)(AA' + BB' + c' + D + E) \\
&\quad (A + BB' + cc' + D + E)(A + BB' + cc' + D' + E) \\
&\quad (AA' + BB' + c + D + E)(AA' + BB' + c + D' + E)
\end{aligned}$$

$$\begin{aligned}
&= (A + BB' + c + D + E)(A + BB' + c + D' + E) \\
&\quad (A + BB' + c + D + E')(A + BB' + c + D' + E') \\
&\quad (AA' + B + c + D + E)(AA' + B' + c + D + E) \\
&\quad (AA' + B + c + D' + E)(AA' + B' + c + D' + E) \\
&\quad (A + BB' + c + D + E)(A + BB' + c' + D + E) \\
&\quad (A + BB' + c + D + E')(A + BB' + c' + D + E') \\
&\quad (AA' + B + c + D + E)(AA' + B' + c + D + E) \\
&\quad (AA' + B + c + D + E')(AA' + B' + c + D + E') \\
&\quad (AA' + B + c + D + E)(AA' + B' + c + D + E) \\
&\quad (AA' + B + c' + D + E)(AA' + B' + c' + D + E) \\
&\quad (A + BB' + c + D + E)(A + BB' + c' + D + E) \\
&\quad (A + BB' + c + D' + E)(A + BB' + c' + D' + E) \\
&\quad (AA' + B + c + D + E)(AA' + B' + c + D + E) \\
&\quad (AA' + B + c + D' + E)(AA' + B' + c + D' + E)
\end{aligned}$$

$$\begin{aligned}
 &= (A+B+C+D+E) (A+B'+C+D+E) (A+B+C+D'+E) (A+B'+C+D'+E) \\
 & (A+B+C+D+E') (A+B'+C+D+E') (A+B+C+D'+E') (A+B'+C+D'+E') \\
 & (A+B+C+D+E) (A'+B+C+D+E) (A+B'+C+D+E) (A'+B'+C+D+E) \\
 & (A+B+C+D'+E) (A'+B+C+D'+E) (A+B'+C+D'+E) (A'+B'+C+D'+E) \\
 & (A+B+C+D+E) (A+B'+C+D+E) (A+B+C'+D+E) (A+B'+C'+D+E) \\
 & (A+B+C+D+E') (A+B'+C+D+E') (A+B+C'+D+E') (A+B'+C'+D+E') \\
 & (A+B+C+D+E) (A'+B+C+D+E) (A+B'+C+D+E) (A'+B'+C+D+E) \\
 & (A+B+C+D+E') (A'+B+C+D+E') (A+B'+C+D+E') (A'+B'+C+D+E') \\
 & (A+B+C+D+E) (A'+B+C+D+E) (A+B'+C+D+E) (A'+B'+C+D+E) \\
 & (A+B+C'+D+E) (A'+B+C'+D+E) (A+B'+C'+D+E) (A'+B'+C'+D+E) \\
 & (A+B+C'+D+E) (A+B'+C'+D+E) (A+B+C'+D+E) (A+B'+C'+D+E) \\
 & (A+B+C+D'+E) (A+B'+C+D'+E) (A+B+C'+D'+E) (A+B'+C'+D'+E) \\
 & (A+B+C+D+E) (A'+B+C+D+E) (A+B'+C+D+E) (A'+B'+C+D+E) \\
 & (A+B+C+D'+E) (A'+B+C+D'+E) (A+B'+C+D'+E) (A'+B'+C+D'+E)
 \end{aligned}$$

$$\begin{aligned}
 &= (A+B+C+D+E) (A+B'+C+D+E) (A+B+C'+D+E) (A+B'+C'+D+E) \\
 & \quad \begin{matrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \end{matrix} \\
 & (A+B+C+D+E') (A+B'+C+D+E') (A+B+C'+D+E') (A+B'+C'+D+E') \\
 & \quad \begin{matrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{matrix} \\
 & (A'+B+C+D+E) (A'+B'+C+D+E) (A'+B+C'+D+E) (A'+B'+C'+D+E) \\
 & \quad \begin{matrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \end{matrix} \\
 & (A+B+C'+D+E) (A+B'+C'+D+E) (A+B+C'+D+E') (A+B'+C'+D+E') \\
 & \quad \begin{matrix} 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \end{matrix} \\
 & (A'+B+C+D+E') (A'+B'+C+D+E') (A'+B+C'+D+E) (A'+B'+C'+D+E) \\
 & \quad \begin{matrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \end{matrix} \\
 & (A+B+C'+D'+E) (A+B'+C'+D'+E) (A'+B+C+D'+E) (A'+B'+C'+D'+E) \\
 & \quad \begin{matrix} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \end{matrix} \\
 & (A'+B+C'+D'+E) (A'+B'+C'+D'+E) \quad \begin{matrix} 1 & 1 & 1 & 1 & 0 \end{matrix}
 \end{aligned}$$

$$= \pi(0, 8, 2, 10, 1, 9, 3, 11, 16, \\ 24, 18, 26, 4, 12, 5, \\ 13, 17, 25, 20, 28, \\ 6, 14, 19, 22, 27, 30)$$

$$= \pi(0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, \\ 13, 14, 16, 17, 18, 19, 20, 22, 24, 25, \\ 26, 27, 28, 30)$$

$$\text{Ans: SOP: } \leq (7, 15, 21, 23, 29, 31)$$

$$\text{pos: } \pi(0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, \\ 13, 14, 16, 17, 18, 19, 20, 22, \\ 24, 25, 26, 27, 28, 30)$$